

ExpressionAuth: Utilizing Avatar Expression Blendshapes for Behavioral Biometrics in VR

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ABSTRACT

As interests in Virtual Reality (VR) continue to rise, head-mounted displays (HMDs) are actively being developed. Current user authentication method in HMDs requires the use of virtual keyboard, which has low usability and is prone to shoulder surfing attacks. This paper introduces ExpressionAuth, a novel authentication method which uses face tracking capabilities in certain HMDs to verify the identity of the user. ExpressionAuth leverages smile as the expression for user verification. ExpressionAuth has the potential to be a secure and usable biometrics, achieving EER as low as 0.00178 and AUC of up to 0.999 in our experiments.

Index Terms: Computing methodologies—Computer graphics—Graphics systems and interfaces—Virtual reality

1 INTRODUCTION

With increasing interests in the Virtual Reality (VR), head-mounted displays (HMDs) are actively being developed. Despite the critical importance of user authentication in HMDs, many issues remain unresolved. Current authentication methods, such as PIN codes, pattern locks, and passwords require the use of handheld controllers, which can pose usability issues, particularly in input where floating virtual keyboards are being used. Additionally, the limited visibility of the surrounding environment when wearing an HMD makes these conventional methods susceptible to shoulder surfing attacks. This vulnerability underscores the need for an authentication method for HMDs that offer both high usability and robust security.

To secure against shoulder surfing attacks, authentication schemes like CueVR have been proposed [1]. However, these knowledge-based authentication schemes depend on user memory, posing a risk of forgotten authentication information. In contrast, biometrics solution, which rely on the user's physical characteristics, offer great convenience as they do not require memorization.

Behavioral biometrics particularly those using body movements, show good promise for HMDs but must mitigate the risk of spoofing attacks [5]. For instance, BlinKey [7], while proposing novel 2-factor authentication method for VR devices, falls under the category of knowledge-based authentications, requiring users to remember user-specific movement patterns. This highlights the ongoing challenge of developing an authentication method for HMDs that is both user-friendly and secure.

Biometrics such as facial and fingerprint recognition focus on an individual's biological characteristics and isn't prone to spoofing attacks. However, they often require external devices, which can impact the usability of HMDs. An ideal HMD authentication method should not only resist third-party observation but also make efficient use of the internal HMD data.

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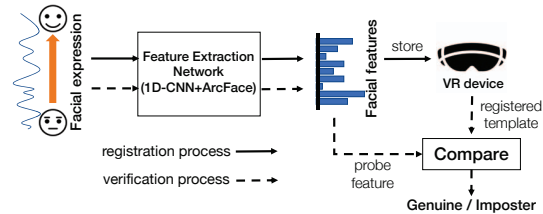


Figure 1: Overview of ExpressionAuth framework

In this paper, we introduce **ExpressionAuth**, a novel biometric authentication method using face tracking to identify HMD users. ExpressionAuth analyzes facial expression via sensors in certain HMDs, extracting individual features with a deep neural network and verifying the user's identity. This approach eliminates the need for external devices and enhances security, as the HMD covers the user's upper face, making it resistant to observation attacks such as shoulder surfing attacks, while maintaining user-friendly authentication scheme.

Regarding identification and psychological effects, research has shown the distinctiveness of smiling in face recognition. For instance, Yacoub et al. demonstrated that smiling faces aid in recognition [6]. We leverage this by capturing the user's smiling expressions as time series data using blendshapes (referred to as "smiling data") from the Meta Movement SDK¹. Smiling data is then processed by a machine learning algorithm to identify users, showcasing the potential of ExpressionAuth for reliable HMD user authentication.

The contributions of this research can be summarized as following three items. (1) We propose ExpressionAuth, a novel biometric authentication which uses face tracking information to identify HMD user. (2) Experiment results on using authentication model trained on a 1D Convolutional Neural Network (1D-CNN) to identify users based on facial expression features. (3) Discussion on ExpressionAuth's usability and robustness against shoulder surfing attacks.

2 EXPRESSIONAUTH

The ExpressionAuth authentication architecture (see Figure 1) consists of enrollment process and verification process. In the enrollment process, the enrollment user is required to take enrollment smiling data. The taken data will be fed into feature extraction network to extract its features. Then the extracted features will be stored in a storage as template features. Note that the storage is typically prepared securely in a VR device. In the verification process, probe smiling data will be taken from a probe user, then feed into the feature extraction network. The extracted probe features is then compared with template features enrolled in the enrollment process by similarity scores calculation. If the similarity score exceeds the threshold, the user will be recognized as genuine user.

¹<https://developer.oculus.com/documentation/unity/move-overview/>

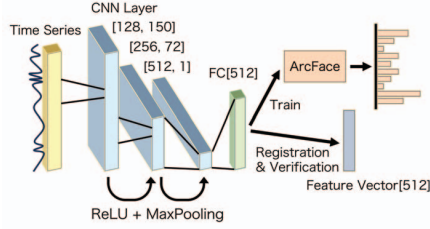


Figure 2: 1D-CNN-based Feature Extraction Network

Blendshapes “Blendshapes” is a simple linear model of facial expressions and is the prevalent approach to realistic facial animation [4]. Our implementation uses Meta Quest Pro HMD, which has face tracking capabilities built-in and can output 63 blendshapes parameters related to facial features.

In this study, we explore the feasibility and accuracy of authentication using these blendshapes, focusing on scenarios where the user’s face is partially obscured by the HMD. By comparing the full set of 63 blendshapes (hereafter, *all-features*) against a subset of 20 blendshapes representing the upper face covered by the HMD (hereafter, *covered-features*).

Machine Learning Model Our model employs a feature-extractive approach commonly used in face recognition, instead of multi-class classification. This method extracts robust features under varying environments and poses, and easily integrates new individuals into the system without retraining. Users first extract their template feature from the smiling data, and during verification we compare similarity scores between template and probe features. The model combines distance learning metrics with a 1D-CNN [3] for authentication.

As shown in Figure 2, our 1D-CNN-based feature extraction network uses ReLU as the activation function and a lightweight architecture of 785,162 parameters and 4.17MB, which is connected to 3 convolutional layers by MaxPooling. Moreover, in order to use the 1D-CNN as a feature extractor, we remove the last full coupling layer and connect the 512-dimensional logit to the distance learning metric ArcFace [2].

3 EXPERIMENTS

A total of 10 participants (age range: 22-25) took part in the data collection. In the data collection process, each participant is required to wear the Meta Quest Pro HMD, press the start button in the data collection application, smile when the first timer (3 seconds) hits 0, then continue to smile until the second timer (2 seconds) ends to take 1 round of data. Data collection is split into 3 sessions with the first session having 50 rounds, the second and third each having 25 rounds, totaling 100 rounds of data for each person. Each person gets labeled 0-9 in the experiment. Each session of data collection is 2 days or more apart.

For accuracy evaluation, two scenarios are considered in our experiment. One is *family scenario*, where the HMD is only used by a limited number of users such as family members. Therefore, the data used to train and test the feature extraction network are from the same people. All 10 classes used to train the feature extraction network are also used in the test data. The other scenario is *public scenario*, where the HMD is used by an unspecified number of users. Therefore, the data used to train and test the feature extraction network are of different persons. 5 classes are used to train the feature extraction network, then another 5 classes are used in test data. To evaluate the accuracy for facial features covered by the HMD. We trained 2 models, one with *all-features*, and the other with *covered-features*. 80% of the smiling data from the first session is used for to train the feature extraction network and the rest is used for evaluation.

Table 1: Performance evaluation of 1D-CNN models trained using *all-features* (63 parameters) and *covered-features* (20 parameters of features covered by the HMD). Session X-Y means that data from session X is being used as template and data from session Y is being used as verification data

	family scenario		public scenario	
	EER ↓	AUC ↑	EER ↓	AUC ↑
<i>all-features</i> (Session1-1)	0.00178	0.999	0.0803	0.981
<i>all-features</i> (Session1-2)	0.0394	0.994	0.122	0.930
<i>all-features</i> (Session1-3)	0.0325	0.994	0.114	0.959
<i>covered-features</i> (Session1-1)	0.0329	0.989	0.178	0.893
<i>covered-features</i> (Session1-2)	0.250	0.810	0.339	0.727
<i>covered-features</i> (Session1-3)	0.192	0.874	0.324	0.736

Table 1 shows the evaluation results of *all-features* and *covered-features* as Error Equal Rate (EER) and Area Under Curve (AUC). In *all-features*, the EER of 0.00178 and AUC of 0.999 for *family scenario* shows that the identification accuracy is high. For *public scenario*, the EER of 0.0803 and AUC of 0.981 shows that the accuracy is slightly lower than the *family scenario*. It can be seen that EER and AUC decreases when we use data from session 2 and 3 as verification data. However, the performance is still high even when data from session 3 is used (*family*: EER = 0.0325 and AUC=0.994, *public*: EER = 0.114 and AUC = 0.959). In *covered-features* the accuracy significantly degrades when session 2 and 3 is used a verification data. The results suggested that we could retain high accuracy by updating the models over time.

4 CONCLUSION AND FUTURE WORK

We proposed a novel biometric authentication scheme which uses facial expression features to authenticate HMD users. Smile is chosen as the expression we use to test our proposal. We collect smiling data from participants then evaluate the effectiveness of the authentication system using 1D-CNN. ExpressionAuth contributes to usable and secure authentication in HMDs, achieving EER as low as 0.00178 and AUC of up to 0.999. The results show that identifying users is possible with smiling data. In the future, we aim to create a continuous biometric authentication scheme where user input is not required.

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